AS Chemistry — Topic 1 Atomic Structure (Exam)

CIE 9701 AS Chemistry · 34 marks · about 40–45 minutes

Topic 1: particles and isotopes, mass spectrometry, electronic configuration / orbitals, and ionisation energy. Answer in English. Section A is Paper 1 style (1 mark each). Section B is written; show working and give the usual CIE reasons (nuclear charge, radius, shielding, spin-pair repulsion) where asked. A Periodic Table may be used.

Section	Content	Marks
A	Multiple choice	13
B1	Iron mass spectrum, particles and ionisation energy	11
B2	Electronic configuration, orbitals, ionisation energy and radii	10
	Total	**34**

Section A — Multiple-choice questions (13 marks)

A1. [1]

The table shows the numbers of protons, neutrons and electrons in four different particles, W, X, Y and Z.

particle	number of protons	number of neutrons	number of electrons
W	32	40	32
X	32	40	34
Y	32	42	32
Z	34	40	34

Which pair represents the atoms of two isotopes of the same element?

A W and Y
B W and Z
C X and Y
D X and Z

A2. [1]

An ion with a charge of 2– contains 10 electrons and 14 neutrons.

What is its nucleon number?

A 14 **B** 22 **C** 24 **D** 26

A3. [1]

What is the total number of protons, neutrons and electrons present in an ammonium ion with a relative formula mass of 21?

	number of protons	number of neutrons	number of electrons
A	11	10	10
B	10	11	11
C	10	11	10
D	11	10	11

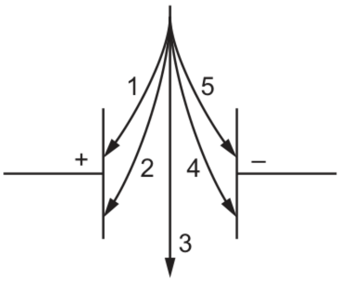
A4. [1]

A helium ion contains two protons, two neutrons and one electron.

This helium ion and a proton are passed separately through a uniform electric field.

The particles are travelling at the same velocity.

Which arrow describes the path of each particle?



	helium ion	proton
A	1	2
B	2	1
C	4	5
D	5	4

A5. [1]

A sample of magnesium contains the isotopes ^{24}Mg, ^{25}Mg and ^{26}Mg only.

The percentage abundance of ^{25}Mg and ^{26}Mg is the same.

The relative atomic mass of magnesium in the sample is 24.3.

What is the percentage abundance of ^{24}Mg?

A 10% **B** 20% **C** 60% **D** 80%

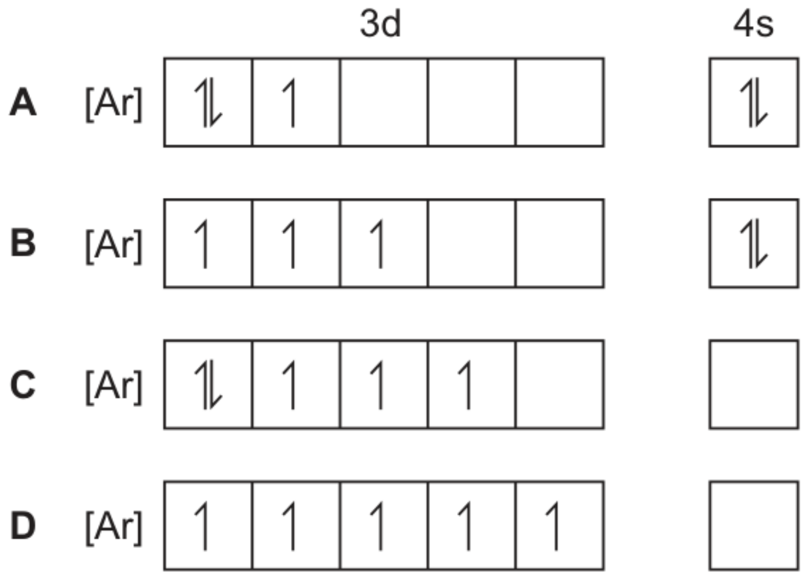
A6. [1]

Which isolated gaseous atom has a total of five electrons occupying spherically shaped orbitals?

A sodium
B fluorine
C boron
D potassium

A7. [1]

What is the electrons in boxes notation for the Fe^{3+} ion?



A8. [1]

Which atom has more unpaired electrons than paired electrons in orbitals of principal quantum number 2?

A carbon
B nitrogen
C oxygen
D fluorine

A9. [1]

Why is the first ionisation energy of oxygen less than that of nitrogen?

A The nitrogen atom has its outer electron in a different subshell.
B The nuclear charge on the oxygen atom is greater than that on the nitrogen atom.
C The oxygen atom has a pair of electrons in one p orbital that repel one another.
D There is more shielding in an oxygen atom.

A10. [1]

Equations involving four enthalpy changes are shown.

$$\begin{aligned}
&Na(g) \longrightarrow Na^{+}(g) + e^{-} && \Delta H = W\\
&Na(g) \longrightarrow Na^{2+}(g) + 2\,e^{-} && \Delta H = X\\
&Na(s) \longrightarrow Na(g) && \Delta H = Y\\
&Na(s) \longrightarrow Na^{2+}(g) + 2\,e^{-} && \Delta H = Z
\end{aligned}$$

Which equation represents the second ionisation energy of sodium?

A X
B $X + Y - W$
C $X - W$
D $Z - W$

A11. [1]

The first seven ionisation energies of an element between lithium and neon in the Periodic Table are shown.

1310 3390 5320 7450 11000 13300 71000 $kJ\,mol^{-1}$

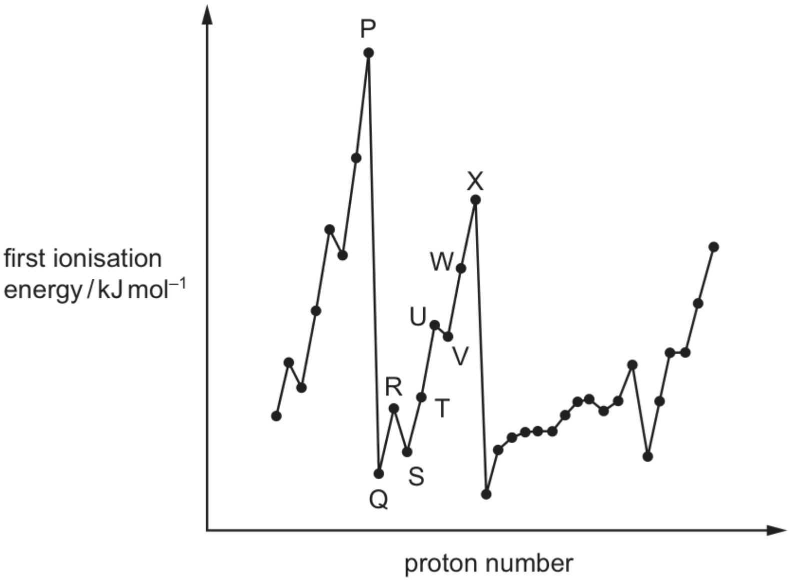
What is the outer electronic configuration of the element?

A $2s^2$
B $2s^2 2p^1$
C $2s^2 2p^4$
D $2s^2 2p^6$

A12. [1]

The graph shows the variation of the first ionisation energy with proton number for some elements.

The letters used are **not** the actual symbols for the elements.



Which statement about the elements is correct?

A P and X are in the same period in the Periodic Table.
B The general increase from Q to X is due to increasing atomic radius.
C The small decrease from R to S is due to decreased shielding.
D The small decrease from U to V is due to repulsion between paired electrons.

A13. [1]

Which statements about first ionisation energies are correct?

1. They are always endothermic.
2. They decrease down Group 2.
3. They decrease across Period 3.

Which statements are correct?	**A**	**B**	**C**	**D**
	1, 2 and 3	1 and 2 only	2 and 3 only	1 only

Section B — Structured questions (21 marks)

B1 · Iron mass spectrum, particles and ionisation energy (11 marks)

A sample of iron is analysed using a mass spectrometer. The mass spectrum shows three isotopes of iron are present in the sample.

B1(a) [1]

Define isotopes.

B1(b)(i) [2]

Fig. 2.1 shows the mass spectrum of the sample of iron.

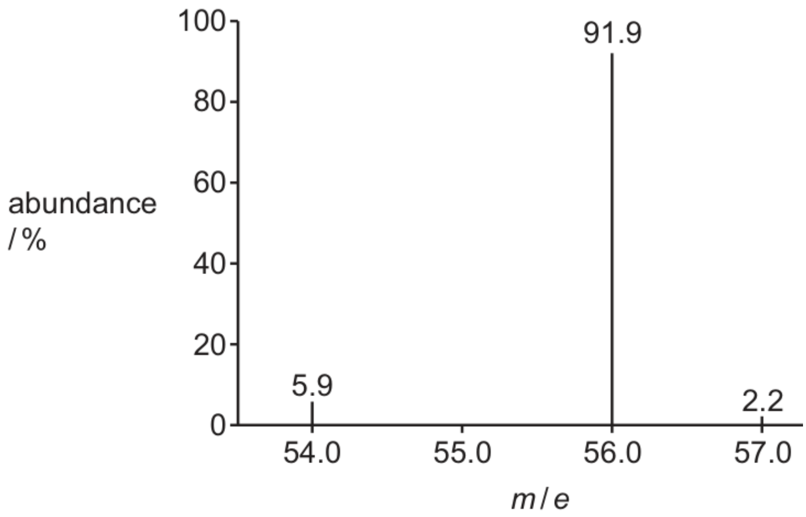


Fig. 2.1

Use Fig. 2.1 to calculate the relative atomic mass, A_r, of iron to one decimal place.
Show your working.

A_r of iron = ____________________

B1(b)(ii) [2]

Complete Table 2.1 to show the number of protons and nucleons in one atom of ^{56}Fe.

particle	number of particles in one atom of ^{56}Fe
protons	
nucleons	

Table 2.1

B1(c) [1]

Deduce the number of pairs of electrons in the shell with principal quantum number $n = 3$ in an Fe atom. number of pairs of electrons ____________________

B1(d) [1]

Write an equation to represent the first ionisation energy of iron.

B1(e) [4]

Suggest how the value for the first ionisation energy of ^{54}Fe compares to the first ionisation energy of ^{56}Fe. Explain your answer in terms of the factors that affect ionisation energy.

B2 · Electronic configuration, orbitals, ionisation energy and radii (10 marks)

The chemical properties of an element are related to the electronic configuration of its atoms.

B2(a)(i) [1]

Give the full electronic configuration of a fluorine atom.

B2(a)(ii) [1]

Deduce the number of pairs of electrons in the second energy shell, $n = 2$, of an oxygen atom. number of pairs of electrons

B2(a)(iii) [1]

Draw the shape of the highest energy orbital that contains electrons in an atom of calcium.

B2(b)(i) [1]

Write an equation to represent the first ionisation energy of sulfur.

B2(b)(ii) [2]

Explain why the first ionisation energy of sulfur is less than the first ionisation energy of phosphorus.

B2(b)(iii) [4]

Arrange the three species F^-, Ne and Na^+ in order of increasing radius.
Explain your answer.

________ < ________ < ________

smallest radius largest radius